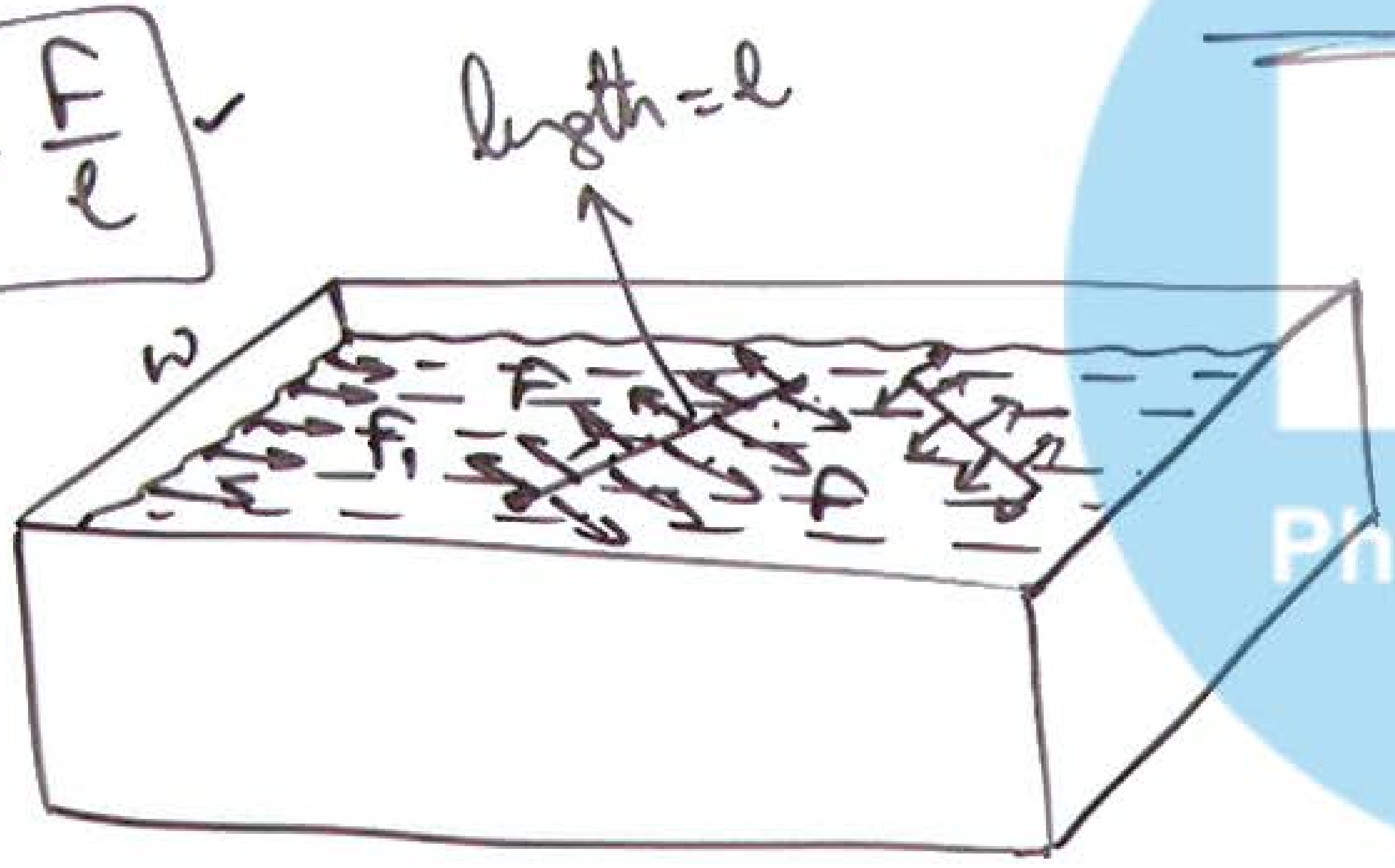
① Understanding Surface Tension:

$$\boxed{T = \frac{F}{l}}$$ ✓



$F_1 = T.w$

At critical temp ✓

$\underline{T = 0}$ ✓

① It is a Scalar

② Decreases with rise in Temperature

③ It depends on impurities

- Insoluble ⇓ $\underline{\text{T decrease.}}$
- Soluble: T may ↑ may ↓

② Concept of Surface Energy:

energy = U

surface area = lw.

$T = \frac{U}{lw}$

$S.E = S.T \times \underline{Area}$

Surface Tension
↓
Surface energy per unit area

drop

$SE = T \cdot 4\pi R^2$

Soap Bubble

$\underbrace{iSE}_{} = \underline{T \cdot 4\pi R^2 \times 2}$

$iSE - fSE = ms\Delta\theta$

Collapse into a drop

$\underbrace{fS.E = T \cdot 4\pi r^2}_{}$

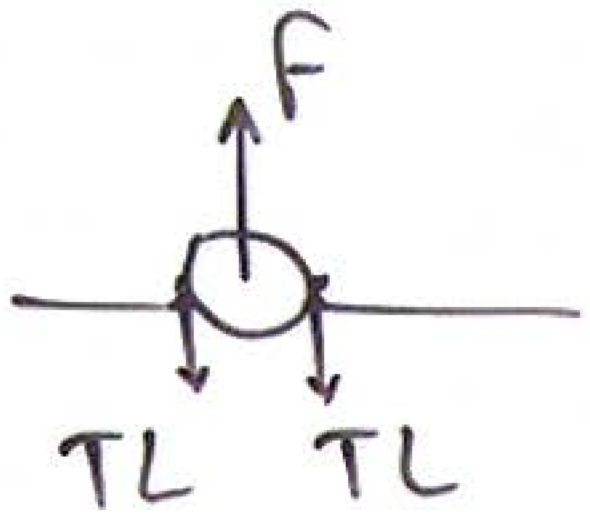
$\frac{4}{3}\pi r^3 = 4\pi R^2 \cdot t$

$r = (3tR^2)^{1/3}$

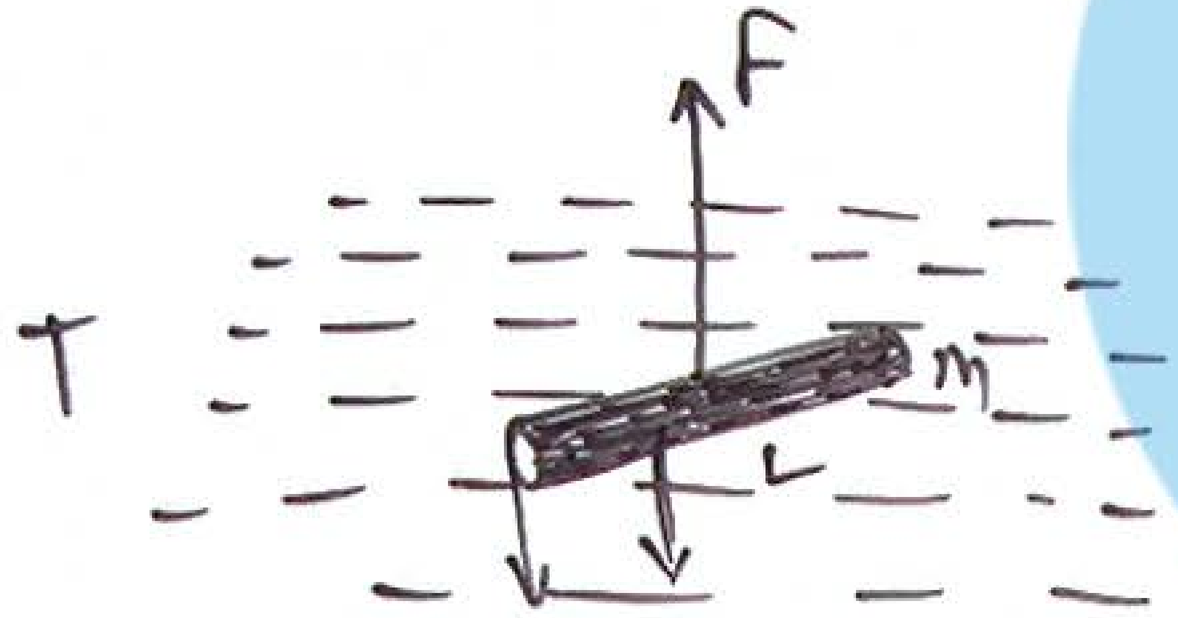
SURFACE TENSION

③ Lifting a floating Body :

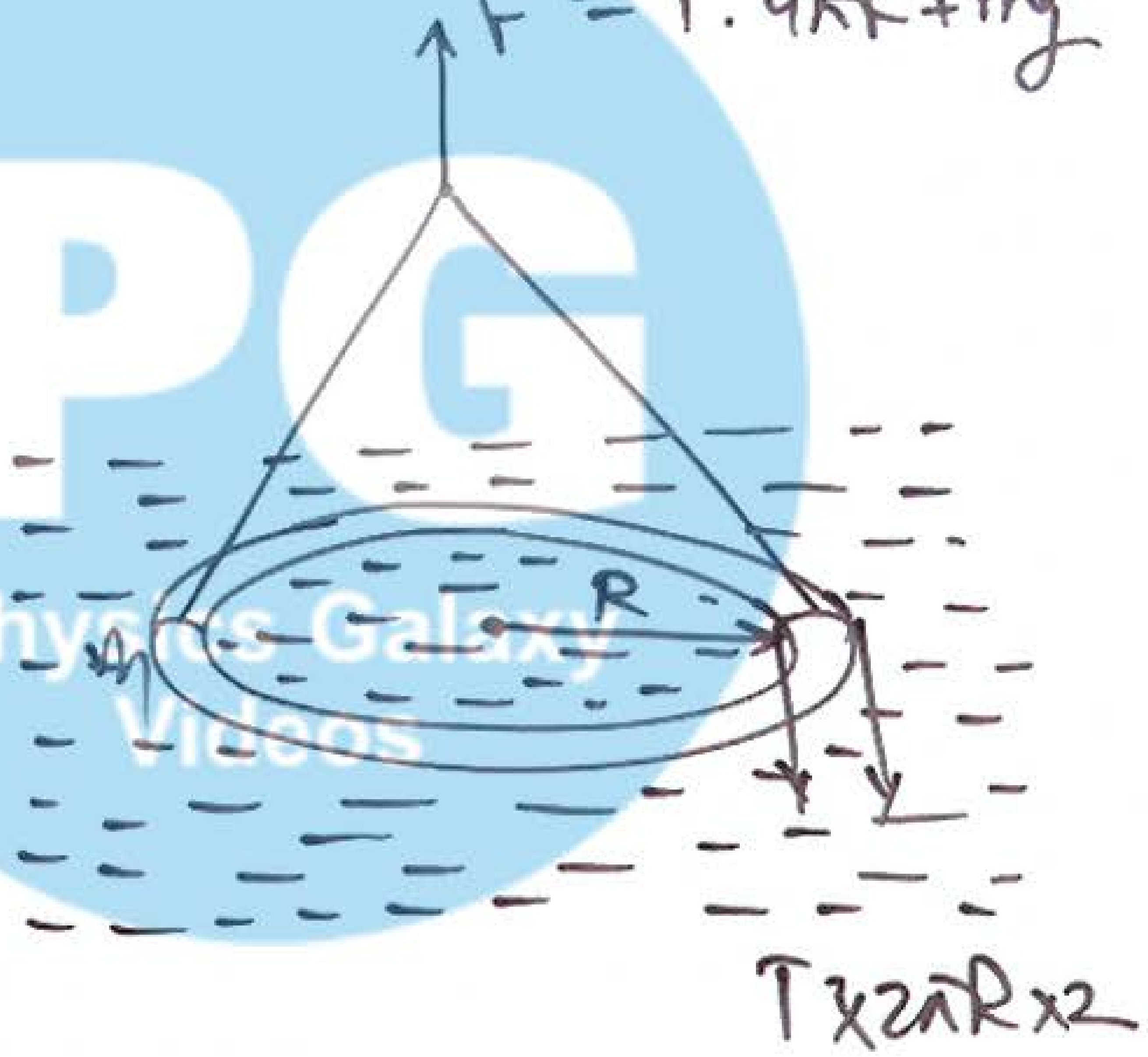
will be considered only if liquid wets the surface of body.



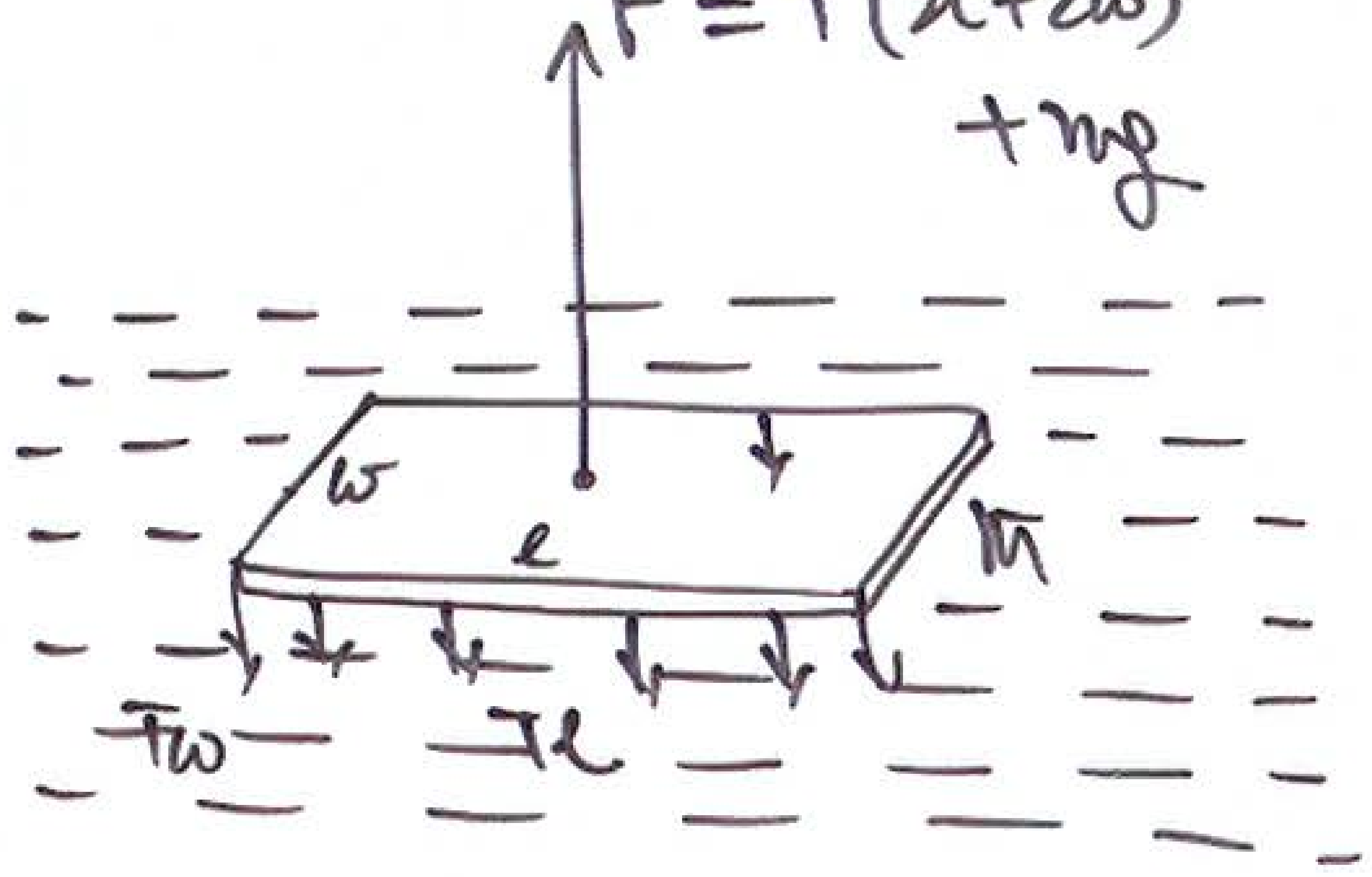
$F = 2TL + mg$



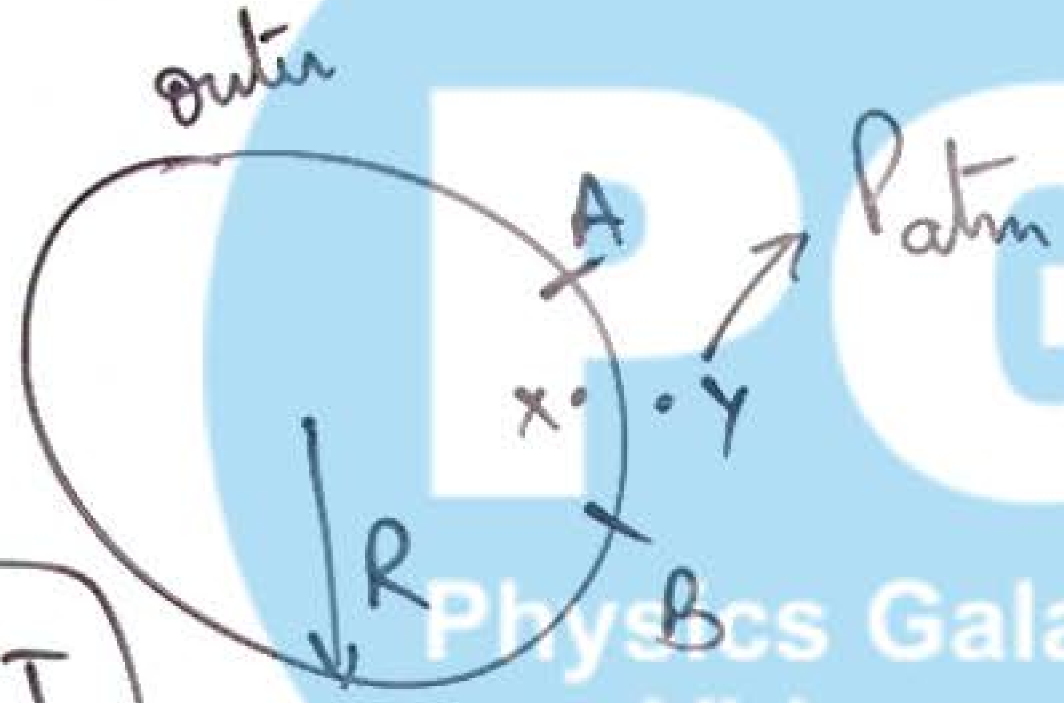
$F = T.4\pi R + mg$



$F = T(2l + 2w) + mg$



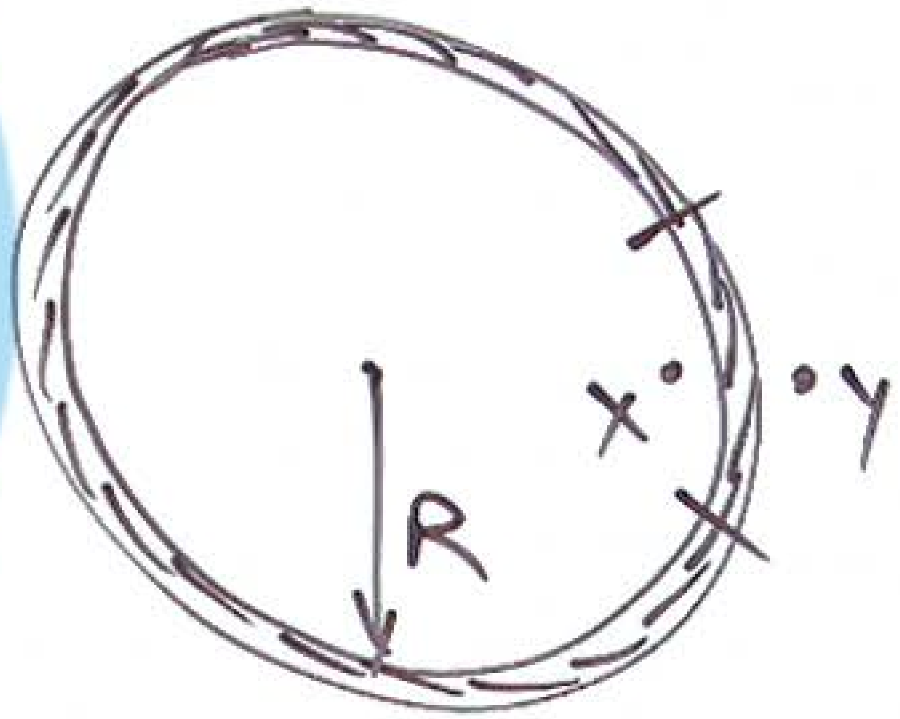
(4) Excess Pressure inside a liquid drop & Bubble:



$$\boxed{P_{ex} = P_x - P_y = \frac{2T}{R}}$$

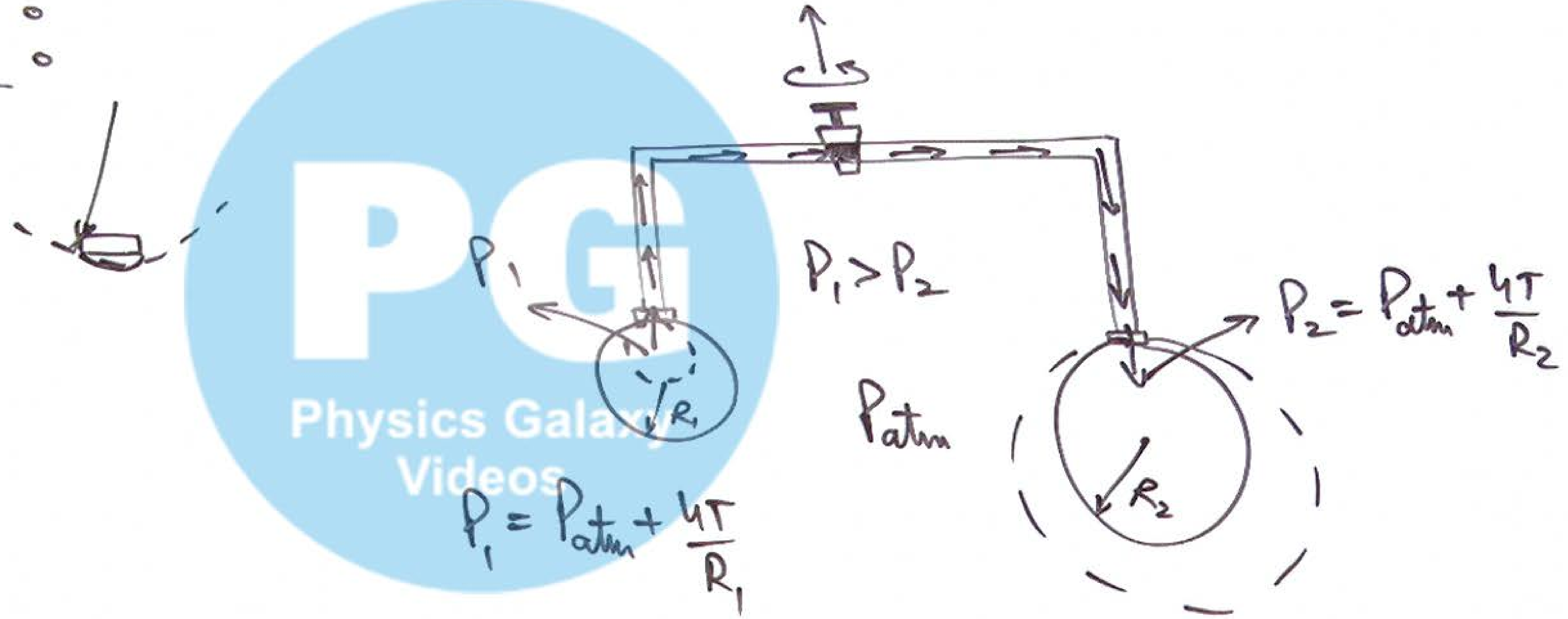
$$P_x > P_y$$

$$P_x = P_{atm} + \frac{2T}{R}$$

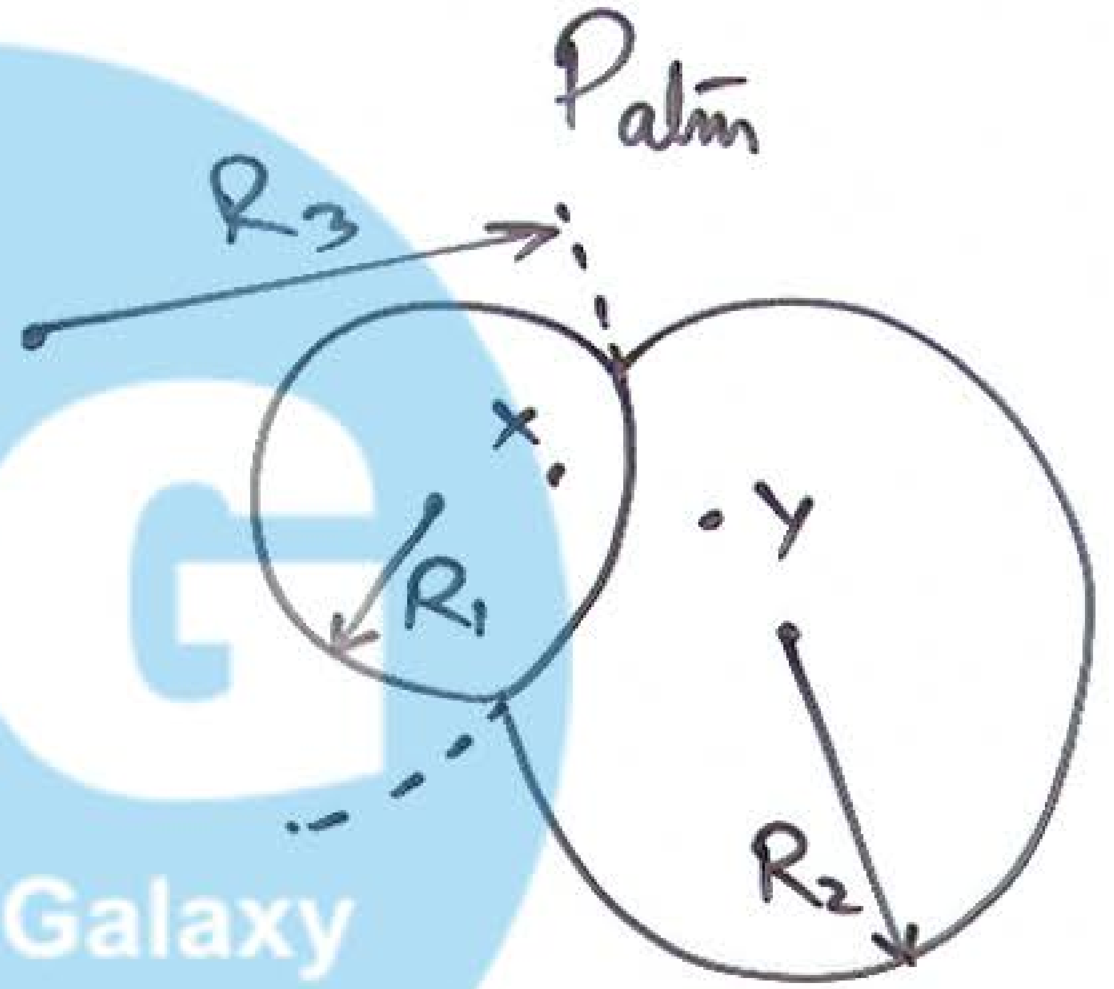


$$\boxed{P_{ex} = P_x - P_y = \frac{4T}{R}}$$

(5) Connecting Bubbles of different Radii :



⑥ Two Soap Bubbles in Contact :



$$P_x = P_{atm} + \frac{4T}{R_1}$$

$$P_y = P_{atm} + \frac{4T}{R_2}$$

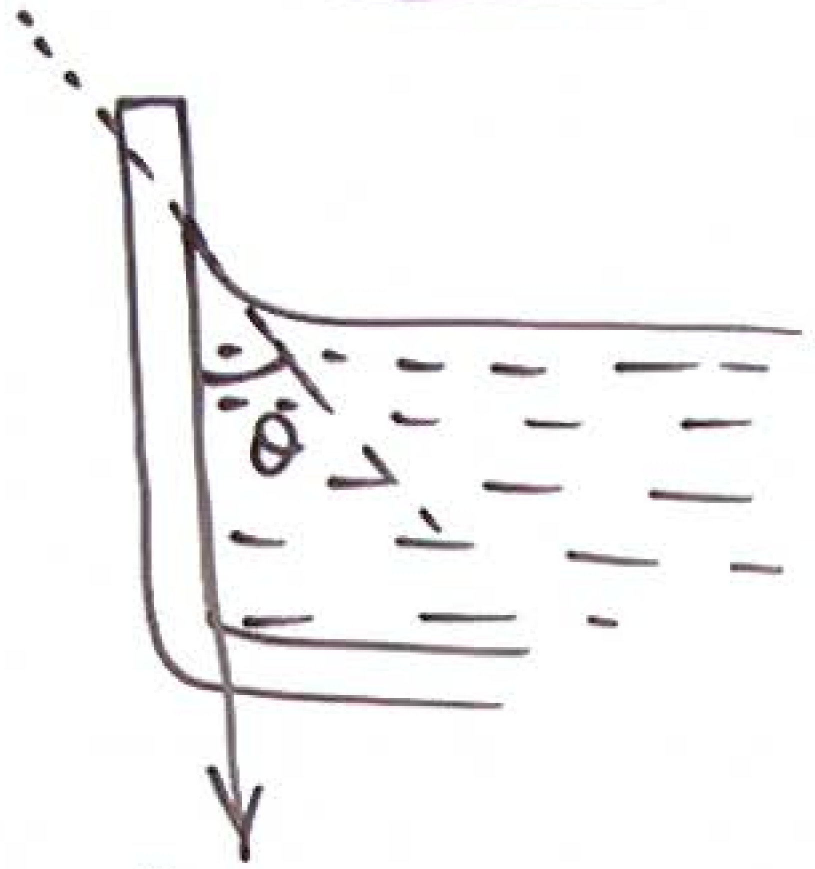
We use

$$P_x - P_y = \frac{4T}{R_3}$$

$$\Rightarrow \quad \cancel{4T}\left[\frac{1}{R_1} - \frac{1}{R_2}\right] = \frac{\cancel{4T}}{R_3} \quad \Rightarrow \quad \boxed{R_3 = \frac{R_1 R_2}{R_2 - R_1}}$$

7 Angle of Contact:

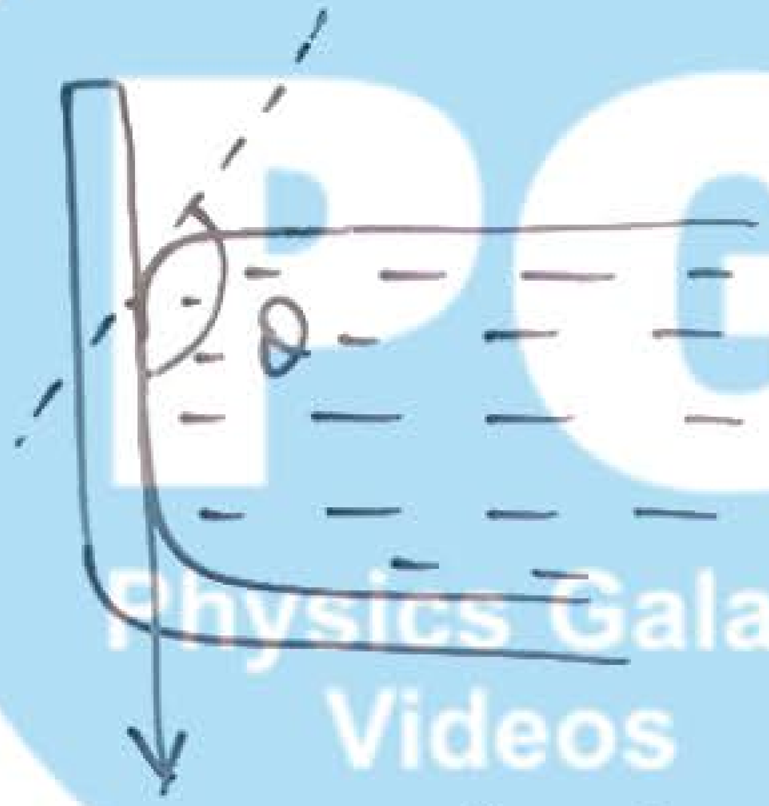
$\theta < 90^\circ$



D.V.

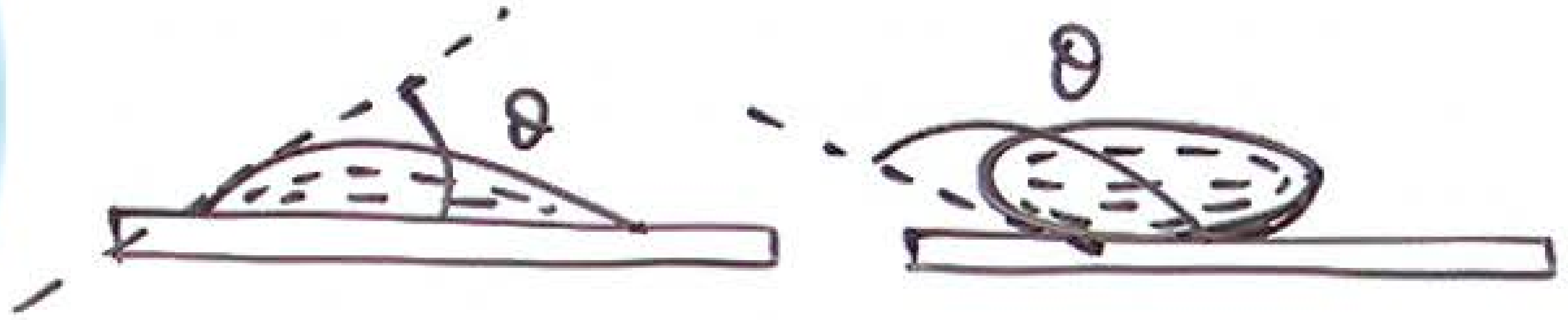
liq wets the solid surface

$\theta > 90^\circ$



D.V.

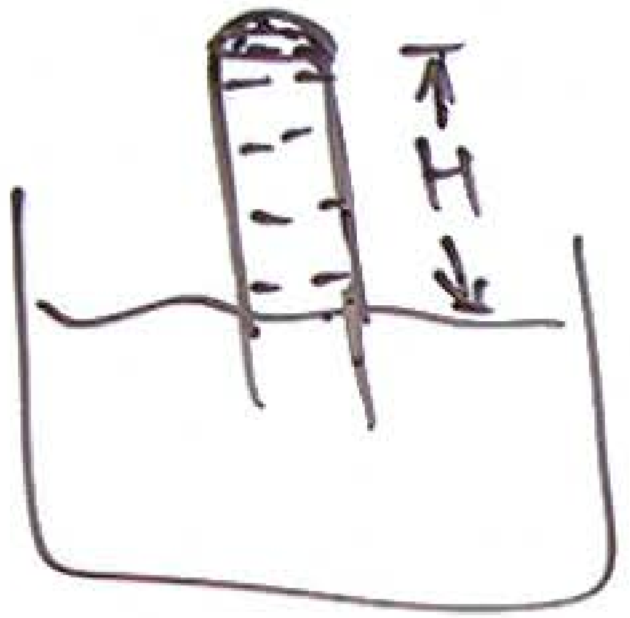
liq does not wet the solid surface.

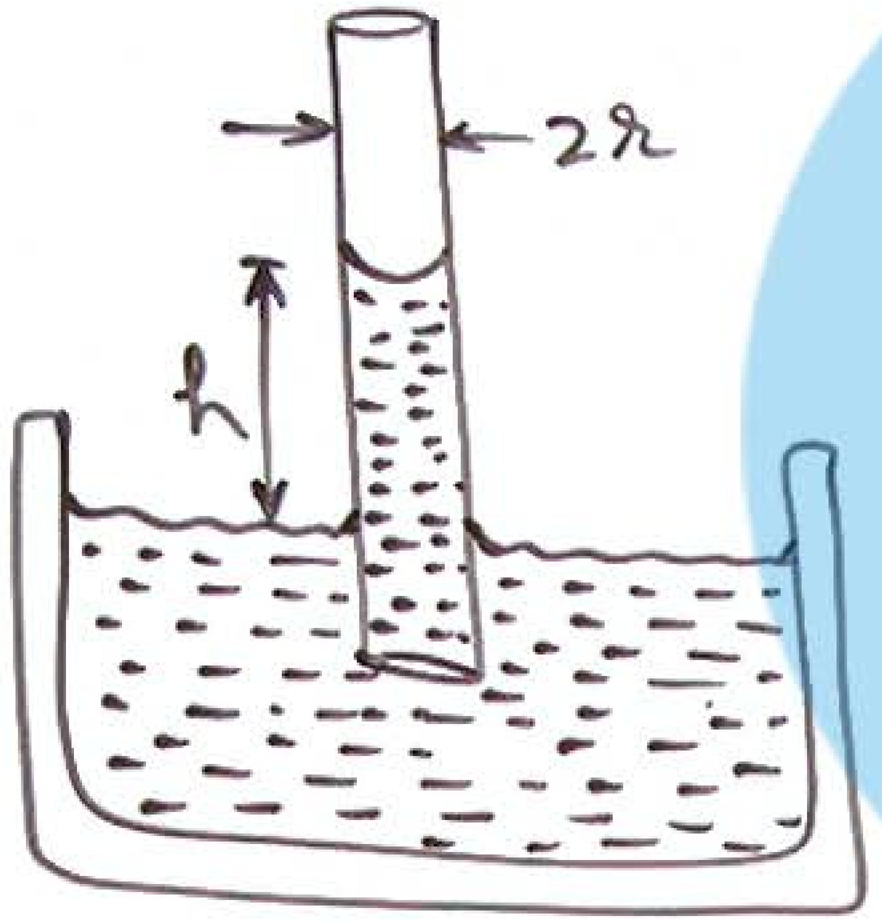


$\theta < 90^\circ$ $\theta > 90^\circ$

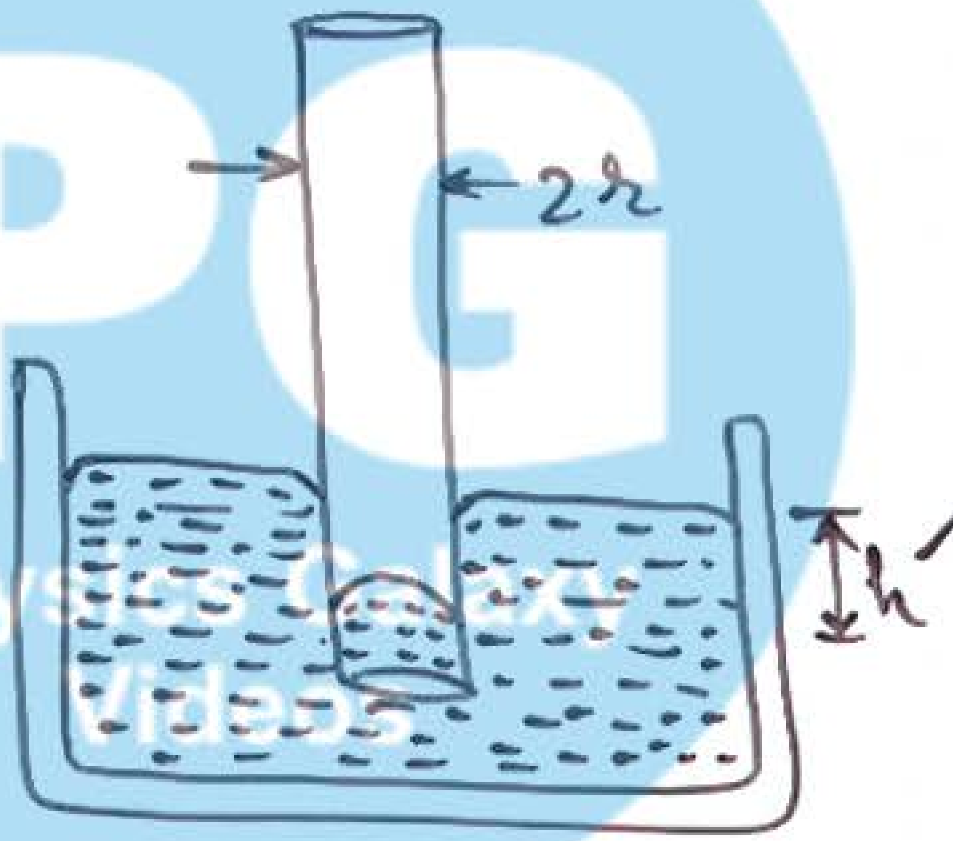
8 Capillary Action:

$H < h$





$\theta < 90^\circ$



$\theta > 90^\circ$

Capillary height

$$\boxed{h = \frac{2T\cos\theta}{r\rho g}} \simeq \frac{2T}{r\rho g}$$

for water glass.

(9) Force required to Separate Two glass plates:

$$\boxed{F = \frac{2TA}{d}}$$ for glass & water

or.

$$\boxed{F = \frac{2TA\cos\theta}{d}}$$